

India, China and the Brahmaputra front

Last week, the Chinese government started work on the construction of a **massive** hydropower project on the Yarlung Zangbo, just before the river **bends** and enters Arunachal Pradesh, where it is called the Siang. The river then flows into Assam, where it is called the Brahmaputra. The \$170 billion **project**, linked to Beijing's developmental goals in the Tibetan region, **is** expected to generate 60 GW of electricity, roughly three times more than the Three Gorges Dam. The project has **stirred** old **anxieties** in the river's **downstream**, especially in Arunachal Pradesh and Bangladesh. Arunachal Chief Minister Pema Khandu has described the Chinese project as a "**ticking water bomb**". An **increase** in the Chinese capacity to **manipulate** the river's flows once the dam is constructed **could** increase the risks of floods in the country's Northeast, he has said. **The fact** that very little is known of the project's storage capacity **has aggravated** the **unease**. The risk could also **stem from** sudden **surges** in downstream flows — from unannounced water releases or from increased **discharge as a result of** the warming climate or even engineering errors. **Hydrology** experts fear that the Chinese project could **disrupt** water flow to the hydro projects **proposed** in the country's Northeast — the region holds nearly half of India's hydropower **potential**, over 80 per cent of which remains **untapped**.

India's Brahmaputra **predicament** is **unlike** that of most lower **riparian** states. The river gains most of its volume only after it enters Arunachal Pradesh, fed by largely **river-fed tributaries** in the Eastern Himalaya, such as the Lohit and the Dibang. In the Brahmaputra's lower riparian regions, the **unpredictability** of flood patterns **is** already a big challenge. The **fears** about **inundation** due to the river's changing flows **are**, therefore, not **unfounded**. In 2013, India and China signed a MoU on sharing information on river flows. But, **by all accounts**, Beijing has not always been open to sharing hydrological data. A more effective **response** to the Chinese dam **would** be to build up the defences of **vulnerable** regions in the Northeast. In 2017, when the Yarlung Zangbo dam was still at a planning stage, Niti Aayog had proposed a multipurpose project in the Siang region as a **strategic counter to** the Chinese hydro station. Besides generating electricity, the project's storage facility can **cushion** the Northeast from the risks of being flooded by water released from the Chinese dam. However, **work** on the Siang dam **has** progressed at a slow **pace**, largely because a section of the local population has opposed the project. The Ministry of Jal Shakti **tasked** the National Hydro Power Corporation to prepare a **pre-feasibility** report but **vital** investigations have not been conducted.

Fears about **displacement** and **livelihood** losses **have** found expression **in the opposition** to the Siang project. The political class and technical experts must come together to build safeguards and remove such **apprehensions**. The **work** on building storage facilities downstream of the Brahmaputra **cannot** be postponed.

[Practice Exercise]

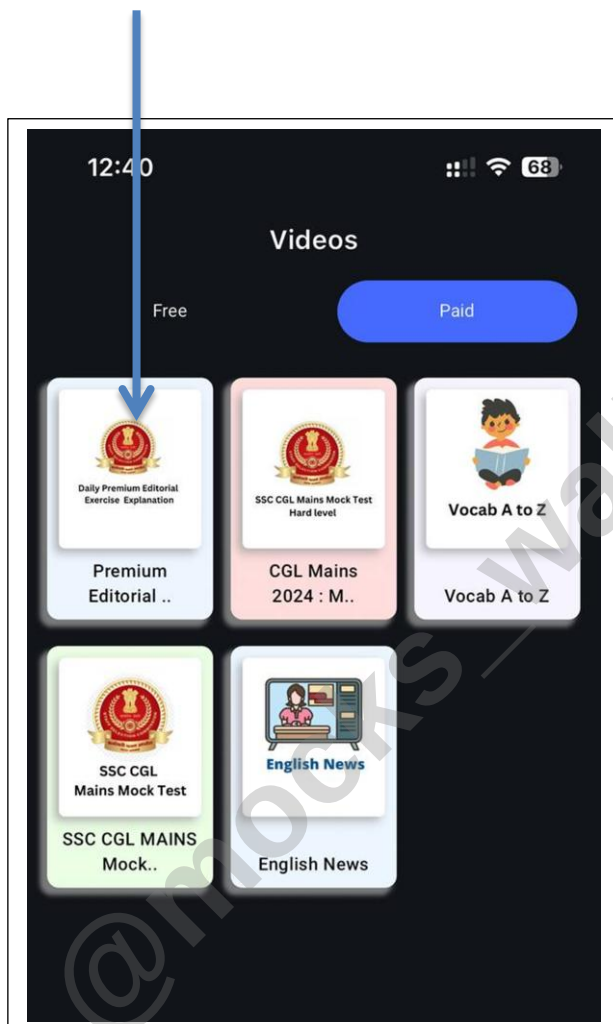
- Red/blue coloring of words in the sentence indicates subject verb relationship; where 'red' denotes 'subject' and 'blue' denotes 'verb'.

Vocabulary

1. **Front** (noun) – Border, boundary, frontier, line, edge सीमा
2. **Massive** (adjective) – Huge, enormous, gigantic, colossal, immense विशाल
3. **Bend** (verb) – Curve, turn, arc, twist, loop मुड़ना
4. **Stir** (verb) – Arouse, provoke, rouse, awaken, ignite उत्तेजित करना
5. **Anxiety** (noun) – Worry, concern, uneasiness, apprehension, fear चिंता
6. **Downstream** (noun) – Lower course, lower stretch, flow end, outflow, river end नदी का निचला हिस्सा
7. **Ticking water bomb** (phrase) – Imminent threat, looming danger, pending hazard, silent risk, potential disaster संभावित खतरा
8. **Manipulate** (verb) – Control, handle, operate, exploit, influence नियंत्रित करना
9. **Aggravate** (verb) – Worsen, intensify, heighten, escalate, inflame और खराब करना
10. **Unease** (noun) – Discomfort, restlessness, anxiety, nervousness, distress बेचैनी
11. **Stem from** (phrasal verb) – Arise from, originate from, result from, emerge from, derive from उत्पन्न होना
12. **Surge** (noun) – Rise, increase, upsurge, escalation, swell वृद्धि
13. **Discharge** (noun) – Outflow, emission, release, effusion, expulsion निकास
14. **As a result of** (phrase) – Because of, due to, owing to, consequent to, on account of के परिणामस्वरूप
15. **Hydrology** (noun) – Water science, water study, hydrodynamics, aquatic science, watershed analysis जल विज्ञान
16. **Disrupt** (verb) – Disturb, interrupt, hamper, derail, destabilize बाधित करना
17. **Propose** (verb) – Suggest, recommend, put forward, offer, advocate प्रस्ताव करना
18. **Potential** (noun) – Capacity, ability, capability, promise, possibility संभाव्यता
19. **Untapped** (adjective) – Unused, undiscovered, unexploited, latent, unutilized अप्रयुक्त
20. **Predicament** (noun) – Dilemma, plight, difficulty, quandary, trouble कठिन स्थिति
21. **Unlike** (preposition) – Dissimilar to, different from, contrary to, opposed to, not like के विपरीत

22. **Riparian** (adjective) – Riverbank-related, streamside, riverside, water-edge, fluvial तटीय
23. **River-fed** (adjective) – Stream-fed, water-supplied, channel-fed, tributary-based, flow-sourced नदी से पोषित
24. **Tributary** (noun) – Stream, branch, feeder, inflow, affluent सहायक नदी
25. **Unpredictability** (noun) – Uncertainty, variability, fluctuation, instability, inconsistency अनिश्चितता
26. **Inundation** (noun) – Flood, deluge, overflow, submergence, torrent बाढ़
27. **Unfounded** (adjective) – Baseless, groundless, unsupported, unproven, unjustified निराधार
28. **By all accounts** (phrase) – Reportedly, according to all sources, from all reports, as said, as per everyone सभी रिपोर्टों के अनुसार
29. **Vulnerable** (adjective) – Exposed, at risk, susceptible, defenseless, unprotected असुरक्षित
30. **Strategic** (adjective) – Tactical, calculated, planned, shrewd, advantageous रणनीतिक
31. **Counter** (to) (noun) – Response, opposition, reaction, check, defense प्रतिकार
32. **Cushion** (verb) – Soften, protect, buffer, shield, absorb आघात को कम करना
33. **Pace** (noun) – Speed, rate, tempo, momentum, velocity गति
34. **Task** (verb) – Assign, delegate, instruct, appoint, direct कार्य सौंपना
35. **Pre-feasibility** (noun) – Preliminary analysis, initial assessment, early study, basic evaluation, scoping study पूर्व-व्यवहार्यता
36. **Vital** (adjective) – Crucial, essential, critical, necessary, indispensable आवश्यक
37. **Displacement** (noun) – Relocation, eviction, uprooting, dislodgment, expulsion विस्थापन
38. **Livelihood** (noun) – Employment, income, occupation, sustenance, means of support आजीविका
39. **In the opposition** (phrase) – Against it, in resistance, objecting to, in disagreement, countering विरोध में
40. **Apprehension** (phrase) – Fear, worry, dread, concern, anxiety आशंका

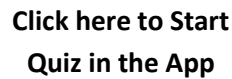
Video solution of Practice Exercise from this Editorial is exclusively available on App.



Summary of the Editorial

1. China has begun construction of a massive hydropower project on the Yarlung Zangbo River, just before it enters Arunachal Pradesh, where it is called the Siang.
2. The river continues into Assam as the Brahmaputra and eventually flows into Bangladesh, making it an international river.
3. The \$170 billion Chinese project is expected to generate 60 GW of power — three times the capacity of the Three Gorges Dam.
4. The project has triggered concerns downstream, particularly in Arunachal Pradesh and Bangladesh, due to potential water flow manipulation.
5. Arunachal Pradesh's Chief Minister Pema Khandu described the Chinese dam as a "ticking water bomb", warning of increased flood risks.
6. There is lack of transparency on the dam's storage capacity, heightening fears of sudden surges and uncontrolled water releases.
7. Hydrology experts worry the dam could affect India's hydropower plans in the Northeast, where 80% of potential remains untapped.
8. India's situation is unique — the Brahmaputra's volume increases significantly after entering Arunachal Pradesh, via tributaries like Lohit and Dibang.
9. The lower riparian regions of the Brahmaputra already face flood unpredictability, making Chinese control over upstream flows especially risky.
10. India and China signed a 2013 MoU for sharing hydrological data, but China has not consistently shared flow information.
11. A more practical response is to strengthen infrastructure in vulnerable areas of Northeast India to mitigate flood risk.
12. In 2017, NITI Aayog proposed a strategic multipurpose dam on the Siang as a counter to China's project.
13. The Siang project could help with power generation and flood control, but progress has been slow due to local opposition.
14. Concerns about displacement and livelihoods have stalled development, despite the Ministry of Jal Shakti assigning NHPC to study feasibility.
15. The editorial urges political leaders and technical experts to work together, address local concerns, and expedite storage facility construction downstream to protect India's interests.

Practice Exercise: SSC Pattern Based



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